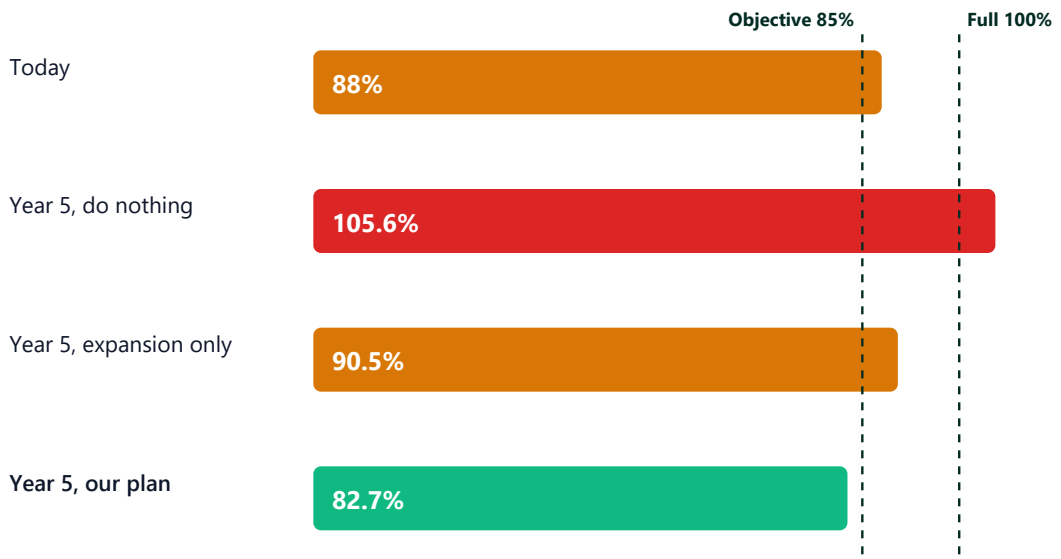
THE CHALLENGE

PortFlow's problem, and why it underperforms

PortFlow's current operating model is struggling to accommodate growing container volumes, resulting in excessive yard congestion, inefficient container movement and delayed truck processing.

88%	6 days	72 min	$14M/yr
YARD UTILIZATION TODAY	AVERAGE DWELL	TRUCK TURN TIME	OVERTIME + PENALTIES
21,120 of 24,000 TEU	every box held 6 days	40 queue + 32 inside	$6M + $8M

YEAR-5 YARD UTILIZATION AT +20% VOLUME



Year-5 objectives: yard utilization <= 85% and truck turn < 45 minutes. Today PortFlow meets neither.

WHY IT UNDERPERFORMS: ONE CAUSE, THREE MECHANISMS

A static yard run on fixed rules and blind data

Blocks are not zoned by demand, stacks ignore pickup order and exact position, the layout never adapts as demand shifts by season, and release status is not shared.

Wrong place
Busy zones sit far from gate, rail and quay exits, so every move travels further — 32 min inside

Wrong order
Stacks ignore pickup order and exact position, so cranes dig and reshuffle — $6M overtime

Wrong time
Seasonal demand shifts but the layout does not; release news arrives late — 6-day dwell

SYMPTOMS MANAGEMENT SEES

Yard crowding	72-min truck turns	$6M overtime
$8M penalties	Customers can't plan	Staff anxiety

THE PROOF, FROM THE CASE DATA

What the case data shows: 21,120 of 24,000 TEU in use today (88%). Add 20% volume at today's 6-day stay and the yard needs about 25,300 TEU, more than it holds. Trucks take 72 minutes against a 45-minute objective, and overtime plus penalties cost $14M a year.

RECOMMENDED PACKAGE

Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot

PILLAR 1

Zone it: demand-based yard zoning (MIP + CRAFT)

Yard zoning and block layout (MIP + CRAFT) — **$1.2M**

MIP assigns container groups to blocks by move frequency and exit; CRAFT swaps zones while flow x distance falls; re-marking and phased moves.

EXPECTED EFFECT

Fast movers next to their exit; time inside 32 to about 22 min

PILLAR 2

Slot it: dynamic container slotting

Dynamic slotting engine with demand forecasting — **$1.6M**

Re-slots containers as demand shifts (e.g. a December peak for one product) and stacks by planned pickup; planners approve.

EXPECTED EFFECT

Stacks follow pickup order and seasonal demand; rehandles -30%

PILLAR 3

Track it: tracking chips and one yard database

Tracking and yard database (digital twin) — **$2.2M**

OCR cameras, RFID tags on vehicles and crane position sensors log every move into one yard database.

Release visibility and truck pickup windows — **$1.4M**

The database shares release and hold status with lines, truckers, rail and customers; trucks book windows for ready boxes.

EXPECTED EFFECT

Exact position of every box; release status and pickup windows shared; dwell 6.0 to 4.7 days

PILLAR 4

Move it: AI AGV pilot (2 units)

AI AGV pilot: 2 units in a fenced lane — **$1.6M**

Two battery AGVs, fleet software linked to the database, navigation, charging and safety fencing; scale only if KPIs pass.

EXPECTED EFFECT

Prove cost per move, uptime and safety before scaling the fleet

ENABLERS

Change management and paid training — **$600K**

Co-design workshops, super-users per shift, AGV supervisor and technician training in paid time.

Contingency (about 9%) — **$800K**

Released at the phase-3 gate if unused.

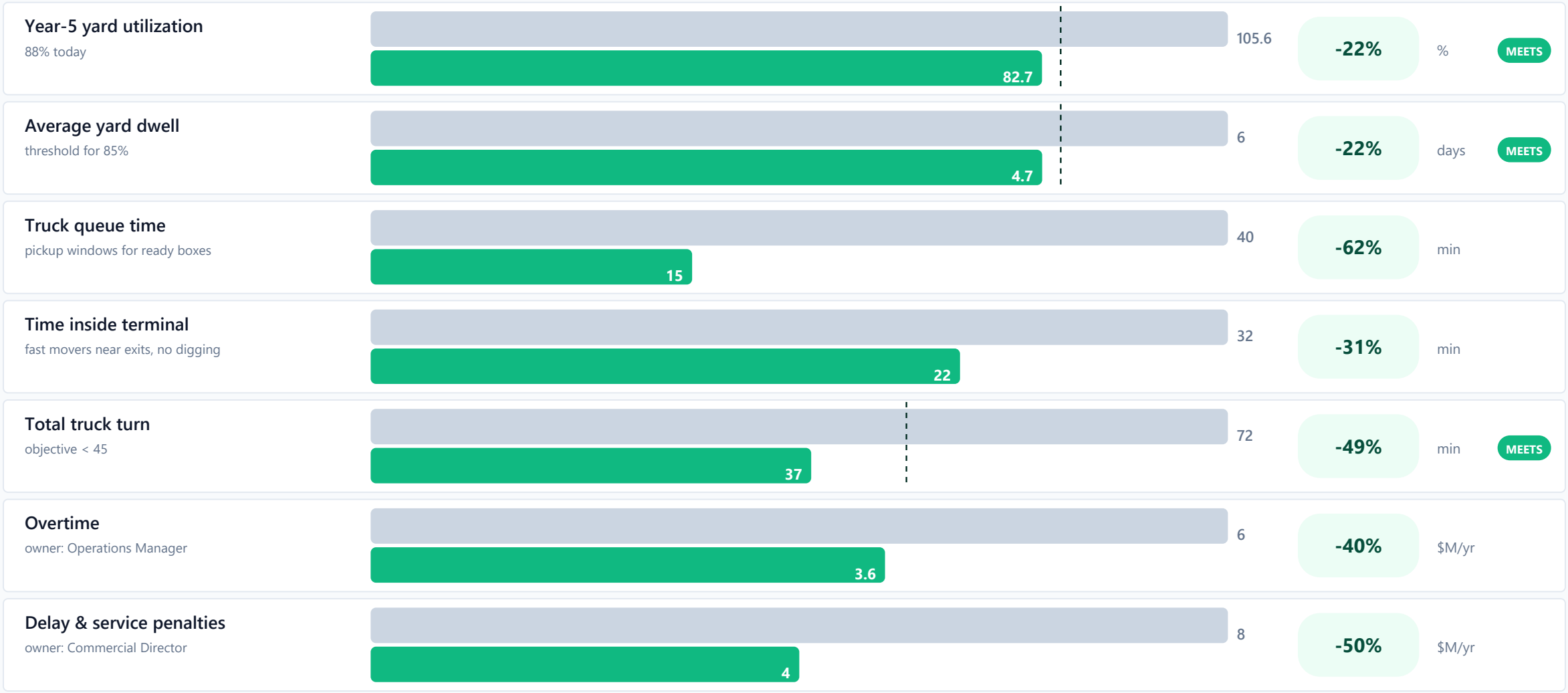
$9.4M of $10.0M upfront · $1.3M/yr to run · every saving has a named owner

Set aside: Expansion only: about $14.5M, over budget, and still 90.5% utilization · Full AGV fleet now: replacing 45 tractors costs far above $10M before a pilot proves it works

KPI IMPACT AT +20% VOLUME

From congestion to flow: every year-5 objective met

Grey = today · green = with the package · dashed line = year-5 objective. All values at +20% volume.



BUDGET ALLOCATION

Budget allocation and decision matrix

WHERE THE $10.0M GOES



Yard zoning and block layout (MIP + CRAFT)	**$1.2M**	12%
Dynamic slotting engine with demand forecasting	**$1.6M**	16%
Tracking and yard database (digital twin)	**$2.2M**	22%
Release visibility and truck pickup windows	**$1.4M**	14%
AI AGV pilot: 2 units in a fenced lane	**$1.6M**	16%
Change management and paid training	**$600K**	6%
Contingency (about 9%)	**$800K**	8%
Total committed	**$9.4M**	
Unspent headroom	$600K	

85% of the spend goes straight at the core issue.
Zone it: $1.2M
Slot it: $1.6M
Track it: $3.6M
Move it: $1.6M
Enablers: $1.4M
Running cost $1.3M a year: software, support, sensors, analysts and refresher training.

DECISION MATRIX (SCORE /10; BUDGET IS A HARD GATE)

	A · Static re-zoning only: one-off MIP and CRAFT layout	B · Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot	C · Yard expansion on the 15-acre parcel with two added RTGs
Target attainment (30%)	1.3	7.4	4.9
Financial return (NPV) (25%)	9.2	10.0	2.0
Payback speed (15%)	6.7	4.6	0.0
Downside resilience (-15% perf.) (15%)	1.7	9.5	6.2
Budget headroom (15%)	8.4	0.6	0.0
Weighted score	5.22	6.92	2.9
Within budget	yes	yes	NO
Meets year-5 objective	no	yes	no
5-year net impact	-$7.4M	-$16.1M	$13.5M

Option B is the only direction that meets the objective inside the budget, and it has the best five-year impact.

FINANCE

The financial case

CASE FORMULA: FIVE-YEAR NET IMPACT = INVESTMENT + 5 X (ADDED ANNUAL COSTS - ANNUAL SAVINGS)

$9.4M + 5 x ($1.3M - $6.4M) = -$16.1M net savings

INVESTMENT VS BUDGET

$9.4M of $10.0M · $600K unspent

Zone it	**$1.2M**	$100K/yr
Slot it	**$1.6M**	$300K/yr
Track it	**$3.6M**	$600K/yr
Move it	**$1.6M**	$150K/yr
Enablers	**$1.4M**	$150K/yr
Total	**$9.4M**	**$1.3M/yr**

ANNUAL SAVINGS, EACH WITH AN OWNER

Overtime -40% — **$2.4M**

$6.0M x 40% · owner: Operations Manager

Shorter travel (zoning), fewer rehandles (slotting), fewer re-plans (one database), smoother truck peaks (pickup windows)

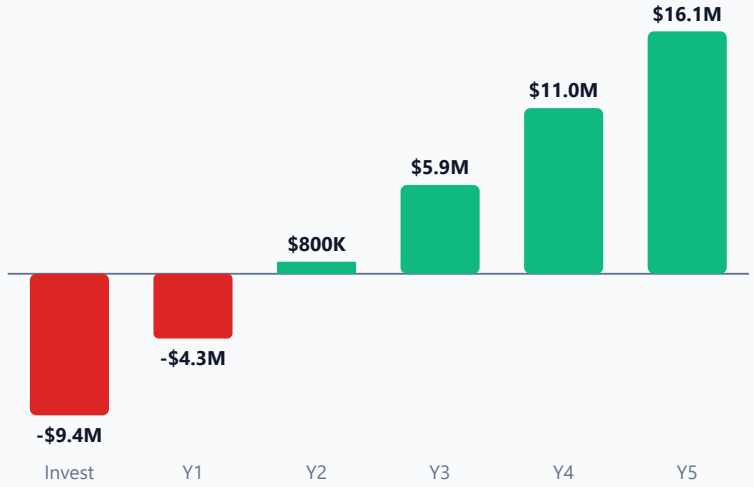
Penalties -50% — **$4.0M**

$8.0M x 50% · owner: Commercial Director

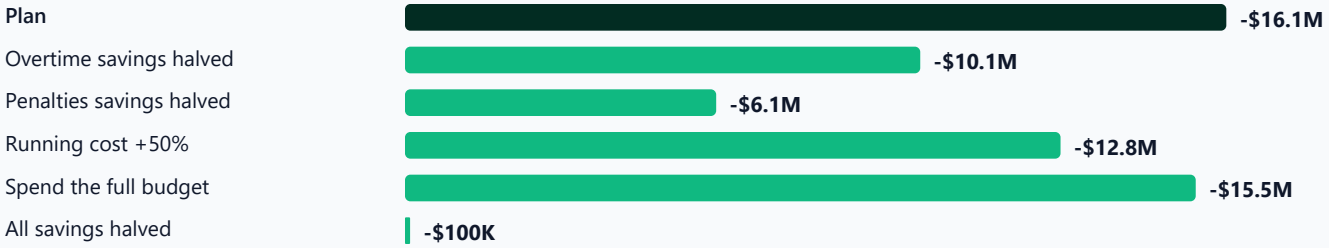
Reliable release status, containers staged before pickup, faster truck service; the other half is assumed to come from customs holds PortFlow cannot override

Savings $6.4M - running cost $1.3M = $5.1M a year

CUMULATIVE POSITION, UNDISCOUNTED



SENSITIVITY, SAME FORMULA



ONE FINANCIAL UNCERTAINTY

How much of the $8M penalty bill comes from customs or documentation holds that PortFlow cannot override. If only a quarter of penalties disappear instead of half, savings drop to $4.4M a year. Break-even needs only $3.2M of savings a year; the investment is paid back in about 22 months of operation.

12 MONTHS, THREE GATED PHASES

Implementing without disrupting operations



HOW OPERATIONS KEEP RUNNING

- Re-zone one block at a time, with housekeeping moves only overnight
- AGVs run only in a fenced pilot lane; manual tractors keep running everywhere else
- Go-lives outside vessel peak windows; sensors fitted on off-peak shifts
- Rollback: if gate throughput drops below baseline for 2 shifts, revert that block or lane

MARKET CONTEXT

Market analysis: what the industry tells us

Volumes are rising, peers compete on reliability, and AI AGVs pay off only when design, maintenance and people line up.

MARKET FORCES

Volumes are growing [1]

The Port of Vancouver handled a record 3.8M TEU in 2025, about 9% above 2024. PortFlow's +20% over five years is in line.

Congestion is the pain point [2]

Container dwell at Canada's largest port was still a significant issue in early 2025, with rail lagging the surge.

Automation is a labour flashpoint [5]

The November 2024 B.C. port lockout centred on automation and manpower; Ottawa ordered binding arbitration.

WHO IS PILOTING AUTONOMOUS VEHICLES

Port of Felixstowe (UK)	Autonomous terminal tractors in mixed traffic since 2022. Believed first in Europe; a small fleet, not a full roll-out.	[7]
Eurogate + Embotech (DE)	6-month autonomous tractor pilot, then a second in Hamburg (2026). Passed 18 test cases: mixed traffic, all weather, cm-level positioning.	[8]

AI AGVS: WHAT THEY CAN AND CANNOT DO [9][10][11]

PROS

+ Runs around the clock with consistent cycle times
+ Battery AGVs need little maintenance (few moving parts)
+ Every move is logged to the yard database
+ Fewer people in crane and truck traffic zones
+ Successful automated terminals cut opex 25-55%

CONS (GO TO THE RISK REGISTER)

- About EUR 150k per unit (2021, conservative) before navigation, charging and fencing
- Needs a fenced lane or certified mixed-traffic safety
- Waiting at crane handoffs can erode productivity
- Fails without preventive maintenance and sensor upkeep
- Some automated terminals lost 7-15% productivity

SO WHAT FOR PORTFLOW

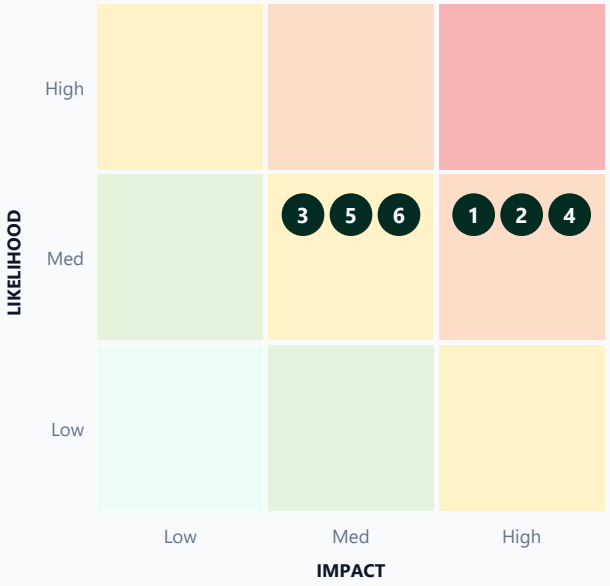
- Pilot 2 AGVs in a fenced lane; scale only if uptime, cost per move and safety beat manual tractors.
- Every AGV weakness becomes a named risk with an owner, budget and early warning (slide 9).
- Pair automation with a no-layoff commitment and retraining to avoid the B.C. dispute.

Sources: [1] Port Technology, Port of Vancouver record 2025 [2] Trains.com, CPKC container dwell at Canada's largest port [5] CBC, B.C. ports frozen amid a labour dispute [7] Maritime Executive, Port of Felixstowe deploys first autonomous trucks [8] Cyprus Shipping News, Embotech and EUROGATE complete 6-month pilot [9] McKinsey, container terminal automation executive summary [10] US DOT ITS cost database, AGV unit cost study [11] TBA Group, AGV vs Lift-AGV vs ALV comparison

EMPLOYEES, AI, PARTNERS, MONEY

Key risks and mitigation

RISK MAP: LIKELIHOOD X IMPACT



Risk budget: $800K contingency held by the Finance Director, plus $600K upfront and $150K/yr for change management and paid training. Each risk has an owner and an early-warning trigger reviewed at the monthly steering meeting.

#	RISK	MITIGATION	OWNER & FUNDING	EARLY WARNING
1	**AGV breakdown or navigation fault** **Medium / High**	Preventive maintenance plan and vendor SLA, sensor health monitoring, fenced lane, manual tractors as instant fallback	Engineering and Maintenance Manager AGV pilot $1.6M + $0.15M/yr maintenance	AGV availability < 90% or any unplanned stop in the lane
2	**Tracking or database errors (wrong slot recorded)** **Medium / High**	OCR and crane-sensor double check on every move, nightly reconciliation audit, manual search procedure kept	IT and Operations Manager Tracking and database $2.2M	Position accuracy < 99% in the daily audit
3	**AI slotting or demand forecast wrong** **Medium / Medium**	Re-slots run weekly on actual bookings; planners approve; segregation rules locked; 6 weeks of shadow mode	Yard Planning Manager Slotting engine $1.6M	Override rate > 30% or rehandles rise for 2 weeks
4	**Re-zoning moves disrupt live operations** **Medium / High**	One block at a time, overnight housekeeping moves only, go-lives outside vessel peaks, rollback plan	Operations Manager Contingency $0.8M	Gate throughput below baseline for 2 shifts
5	**Employee resistance to AGVs and AI** **Medium / Medium**	No layoffs from this program; drivers retrain as AGV supervisors and technicians; staff co-design before tools arrive	HR Director with the Employee Representative Change and training $0.6M + $0.15M/yr	Training completion < 90% or advice acceptance < 50%
6	**AGV pilot costs more than it saves** **Medium / Medium**	Fixed-price quote, pilot capped at 2 units, fleet expands only if the month-12 gate passes	Finance Director No extra spend until the gate	Cost per move above manual tractor after 3 months